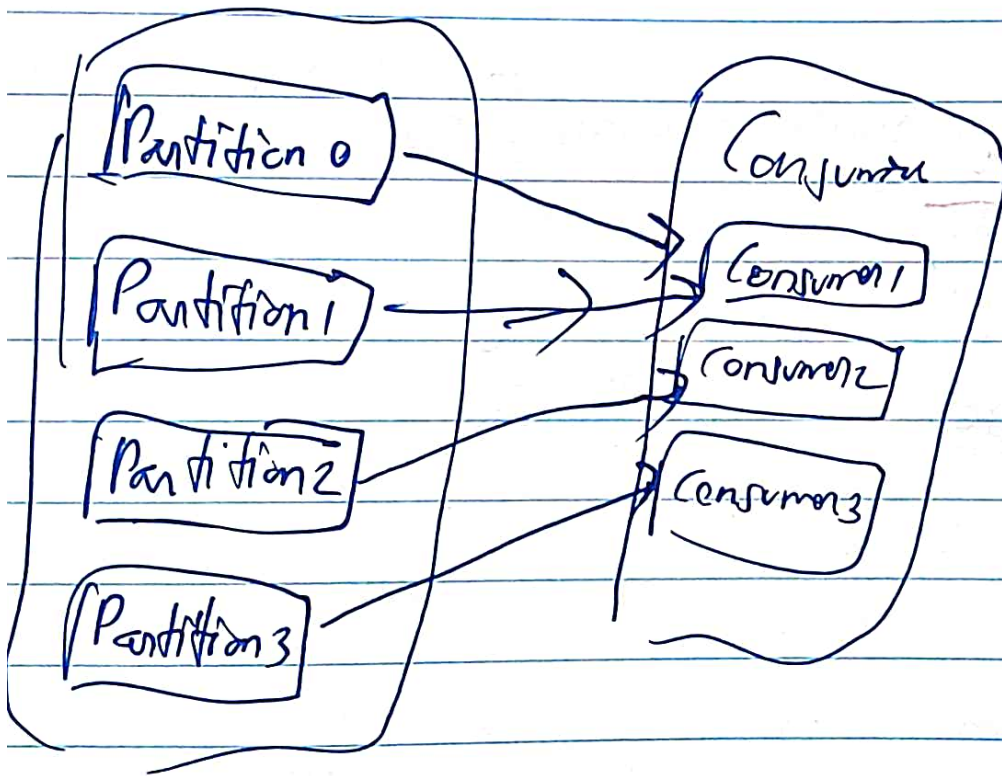


~~(Chapter 5 Partitioning)~~

(Chapter 4 Kafka Consumers)

Consumer group subscribed to a topic.
Each topic would accordingly ~~send msg~~
assign partitions to consumers in the group.



~~Box~~ Multiple consumer group can
subscribe to a single topic.

new partition gets added or consumer is dropped, rebalancing is triggered.

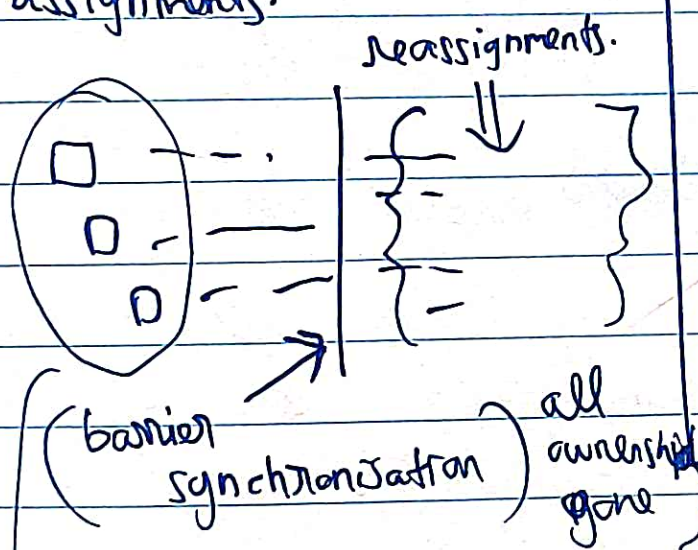
2 types of rebalancing.

{ group coordinator → membership
→ heartbeat }

Eager.

Cooperative.

all consumers give up ownership of partition and re-join with brand new assignments.



group coordinator, (owned whole consumer group or leader maintains the assignment mapping).

Happens in 2 phases

1) Group coordinator, tells few partitions and consumers will be reassigned. those stop consuming

2). Group leader assigns new partition to their consumers. Takes few iterations to stabilize.

(But does not become unavailable like eager rebalancing)

Note: For a consumer group.

Leader, Group Coordinator are different.

- First consumer to join group is leader.
- Leader responsible for assigning partitions to consumer groups.
- Leader sending it to group coordinator.
- Group coordinator sending it to all consumers.

(Poll Loop)

→ consumer.poll(timeout);

1st time .poll() call responsible to ~~call~~ find group coordinator and ~~then~~ get assigned partitions for it and do the work.

(max.interval.poll.ms) → tells how much time can a consumer be considered alive without making a poll call.

auto.offset.reset → let's say a consumer comes back after being dead, let's say grp-coordinator is unable to find which offset it's supposed to be part of from again.

Options

- Latest → end of partition
- Earliest → beginning of partition
- None → error.

enable.auto.commit → to commit offset to it automatically

two types of offset commit can be done to broker.

```
commitSync();  
commitAsync();
```

 →

Since we poll batch of messages so last offset of the batch's offset is sent.

For each map have a map of

$\text{Map}(\text{TopicPartition}, \text{offsetMetadataRecord})$